

```

create database db1;
use db1;

create table company(
                                comp_id int primary key,
    company_name varchar(50),
    city varchar(50)
);

create table employee(
                                emp_id int primary key,
    employee_name varchar(50),
    street varchar(50),
    city varchar(50),
    email_id varchar(50),
    comp_id int,
                                foreign key(comp_id) references
company(comp_id),
    salary decimal
);

```

```

create table manages(
                                emp_id int primary key,
    mgr_id int,
    foreign key(emp_id) references employee (emp_id),
    foreign key(mgr_id) references employee (emp_id)
);

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```

INSERT INTO company (comp_id, company_name, city) VALUES
(101, 'First Bank Corporation', 'New Delhi'),
(102, 'Yes Bank Corporation', 'Mumbai'),
(103, 'Small Bank Corporation', 'Varanasi'),
(104, 'Tech Solutions Inc', 'Bangalore');

```

```

INSERT INTO employee (emp_id, employee_name, street, city, email_id, comp_id,
salary) VALUES
(1, 'Smith Jones', 'Park Street', 'New Delhi', 'smith@fbc.com', 101, 65000),
(2, 'Priya Sharma', 'Ghat Road', 'Varanasi', 'priya@sbc.com', 103, 45000),
(3, 'Amit Patel', 'MG Road', 'Mumbai', 'amit@ybc.com', 102, 35000),
(4, 'Rahul Verma', 'Link Road', 'New Delhi', 'rahul@fbc.com', 101, 12000),
(5, 'Sneha Rao', 'Indira Nagar', 'Bangalore', 'sneha@ts.com', 104, 55000),
(6, 'John Doe', 'Church Street', 'Mumbai', 'john@ybc.com', 102, 25000),
(7, 'Pankaj Kumar', 'Assi Ghat', 'Varanasi', 'pankaj@fbc.com', 101, 9000),
(8, 'Anjali Singh', 'Civil Lines', 'New Delhi', 'anjali@fbc.com', 101, 48000),
(9, 'Vikram Seth', 'Main Street', 'Mumbai', 'vikram@ybc.com', 102, 32000),
(10, 'Pooja Hegde', 'Ring Road', 'Varanasi', 'pooja@sbc.com', 103, 21000),
(11, 'David Miller', 'Baker St', 'New Delhi', 'david@fbc.com', 101, 70000),
(12, 'Suresh Raina', 'Lanka', 'Varanasi', 'suresh@sbc.com', 103, 15000),

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(13, 'Preeti Zinta', 'Film City', 'Mumbai', 'preeti@ybc.com', 102, 52000),
(14, 'Robert Frost', 'Poet Lane', 'Bangalore', 'robert@ts.com', 104, 41000),
(15, 'Alice Cooper', 'Rock St', 'New Delhi', 'alice@fbc.com', 101, 28000),
(16, 'Peter Parker', 'Web Road', 'Mumbai', 'peter@ybc.com', 102, 19000),
(17, 'Karan Johar', 'Bandra', 'Mumbai', 'karan@ybc.com', 102, 85000),
(18, 'Prakash Raj', 'Cantonment', 'Varanasi', 'prakash@sbc.com', 103, 33000),
(19, 'Mohit Suri', 'Ghat Road', 'Varanasi', 'mohit@sbc.com', 103, 22000),
(20, 'Sara Ali', 'New Colony', 'New Delhi', 'sara@fbc.com', 101, 11000);
```

```
INSERT INTO manages (emp_id, mgr_id) VALUES
```

```
(4, 1), (8, 1), (11, 1), (15, 1), (20, 1), -- Smith Jones (1) manages several at
FBC
(7, 2), (10, 2), (12, 18), (19, 18),      -- Priya (2) and Prakash (18) manage
others
(3, 17), (6, 17), (9, 13), (16, 13);      -- Management at Yes Bank
```

```
-- Find the names and cities of residence of all employees who work for First Bank
Corporation.
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```
select e.employee_name, e.city
from employee e join company c
on e.comp_id=c.comp_id
where c.company_name = 'First Bank Corporation';
```

```
-- Find the names of all employees who either work for First Bank Corporation or
work for Yes Bank Corporation
```

```
select e.employee_name from
employee e join company c
on e.comp_id=c.comp_id
where c.company_name='First Bank Corporation'
OR c.company_name = 'Yes Bank Corporation';
```

```
-- Find the names of employees and their company-name who take a salary of between
Rs. 20000 and Rs. 50000.
```

```
select e.employee_name, c.company_name from
employee e join company c
on e.comp_id=c.comp_id
where salary between 20000 and 50000;
```

```
-- Find the names, street addresses, and cities of residence of all employees who
work for First Bank Corporation and earn more than Rs. 10,000.
```

```
select e.employee_name, e.street, e.city from
employee e join company c
on e.comp_id = c.comp_id
where c.company_name='First Bank Corporation'
and e.salary>10000;
```

```
-- Find all employees in the database who do not work for First Bank Corporation.
```

```
select * from employee e join company c
on e.comp_id=c.comp_id
```

```

where c.company_name != 'First Bank Corporation';

-- Find the names of all employees who work under the manager Smith Jones
select e.employee_name from
employee e join manages m
on e.emp_id=m.emp_id
join employee mgr on m.mgr_id=mgr.emp_id
where mgr.employee_name='Smith Jones';

-- List all the manager names who lives in city Varanasi
select distinct mgr.employee_name
from employee mgr where mgr.emp_id in
(select mgr_id from manages)
and mgr.city='Varanasi';

-- Find the Name and city of residence and salary of all the managers whose name
starts with P.
SELECT DISTINCT e.employee_name, e.city, e.salary
FROM employee e
JOIN manages m ON e.emp_id = m.mgr_id
WHERE e.employee_name LIKE 'P%';

-- Find the name of the employee whose Company name's 3rd last character is 'i'.
SELECT e.employee_name
FROM employee e
JOIN company c ON e.comp_id = c.comp_id
WHERE c.company_name LIKE '%i__';

-- Find all employees in the database who live in the same cities as the companies
for which they work
SELECT e.*
FROM employee e
JOIN company c ON e.comp_id = c.comp_id
WHERE e.city = c.city;

-- Find all employees in the database who live in the same cities and on the same
streets as do their managers
SELECT e.*
FROM employee e
JOIN manages m ON e.emp_id = m.emp_id
JOIN employee mgr ON m.mgr_id = mgr.emp_id
WHERE e.street = mgr.street AND e.city = mgr.city;

-- Find all employees in the database who earn more than each employee of Small
Bank Corporation.
SELECT * FROM employee
WHERE salary > ALL (
    SELECT e.salary FROM employee e
    JOIN company c ON e.comp_id = c.comp_id
    WHERE c.company_name = 'Small Bank Corporation'

```

```

);

-- Find all employees who earn more than the average salary of all employees of
their company.
SELECT * FROM employee e1
WHERE salary > (
    SELECT AVG(salary) FROM employee e2
    WHERE e1.comp_id = e2.comp_id
);

-- Find the company that has the most employees.
SELECT c.company_name
FROM company c
JOIN employee e ON c.comp_id = e.comp_id
GROUP BY c.company_name
ORDER BY COUNT(e.emp_id) DESC limit 1;

-- Find the company that has the smallest payroll.
SELECT c.company_name
FROM company c
JOIN employee e ON c.comp_id = e.comp_id
GROUP BY c.company_name
ORDER BY SUM(e.salary) ASC limit 1;

-- Find those companies whose employees earn a higher salary, on average, than the
average salary at First Bank Corporation.
SELECT c.company_name
FROM company c
JOIN employee e ON c.comp_id = e.comp_id
GROUP BY c.company_name
HAVING AVG(e.salary) > (
    SELECT AVG(e2.salary) FROM employee e2
    JOIN company c2 ON e2.comp_id = c2.comp_id
    WHERE c2.company_name = 'First Bank Corporation'
);

-- Modify the database so that Jones now lives in New Delhi
SET SQL_SAFE_UPDATES = 0;
UPDATE employee
SET city = 'New Delhi'
WHERE employee_name LIKE '%Jones';

-- Give all employees of First Bank Corporation a 10 percent raise in their salary
UPDATE employee
SET salary = salary * 1.10
WHERE comp_id = (SELECT comp_id FROM company WHERE company_name = 'First Bank
Corporation');

```